PODIUM — the reading list

Six things worth getting good at, where to learn them, and what to write down

16 October 2026

This is for the notepad, not the laptop. Six topics, ranked by how much each one unblocks. Each has one thing to start with, a few to go deeper, and — the part that matters — **specific questions to answer in your own handwriting.**

A note on why the questions matter more than the reading. You will forget 90% of what you read this week. You will not forget what you wrote down in your own words, because writing it forces you to find out whether you actually understood it. If you can't write the answer, you haven't learned it yet, and that is useful information rather than a failure.

	Topic	Why it is where it is
1	Asking questions without selling	The gym owner conversation is the only thing the whole business is waiting on
2	How a gym actually makes money	You are asking for R1,500 a month. You need to know what that is next to
3	One light	The product is a photograph. This is the craft of it
4	Distance, and what it does to a body	You just measured it. Now understand what you measured
5	Reading a number honestly	You have the instincts already. This gives them names you can defend
6	OpenSCAD	Because you are doing the pod yourself

How to take the notes

One page per source. Four headings, every time:

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SOURCE                  (what it was, and the date you read it)

WHAT IT SAID            three to five bullets, in your own words.
                        If you are copying phrases, you are not
                        learning, you are transcribing.

WHAT IT CHANGES ABOUT   the bit that matters. If nothing — write
PODIUM                  "nothing" and say why. That is a real answer.

WHAT I STILL DON'T KNOW the question it raised and did not answer.
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Then keep **one running page at the back** titled *Questions I cannot answer yet*. Add to it constantly, cross things off as you find out. In six weeks that page is the most valuable thing in the notepad — it is the map of what you don't know, and nobody else can write it for you.

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1 · Asking questions without selling

Why this first. Everything is waiting on one conversation with the Aspire owner, and the most likely way to waste it is to walk in and pitch. A pitch gets you politeness. Politeness tells you nothing, and you will come away feeling good and knowing exactly as much as you did before.

Start here: *The Mom Test* by Rob Fitzpatrick. It is 118 pages and you can read it in an evening. The whole idea is one sentence: **ask about what people have already done, not what they think they would do.** "Would you pay R1,500 a month for this?" is a question your mother would say yes to. "What did you pay for the last thing you bought to keep members?" cannot be answered politely.

Free summaries if the book has to wait: a neutral [book report at mtlynch.io](https://mtlynch.io/book-reports/the-mom-test/), and [a walkthrough of the question types](https://usercall.co/post/the-mom-test-customer-interview-questions). Read the book anyway — the summaries give you the rule and not the practice.

Write in the notepad:

1. Rewrite all six of your gym-owner questions so that each asks about something that has **already happened**. Which ones survived unchanged?
2. What is the single question you most want to ask, and what is the honest version of it that cannot be answered with encouragement?
3. Write down three compliments you would be pleased to hear from the owner — then, next to each, what it actually proves. (Usually: nothing.)
4. What would the owner have to **do**, not say, for you to believe they are interested? Book a date? Introduce you to someone? Give you floor space?

You've got it when you catch yourself about to ask a hypothetical and change the question mid-sentence.

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2 · How a gym actually makes money

Why. You are about to ask a business owner for R1,500 a month. They will not evaluate that against your costs — they will evaluate it against what else R1,500 buys them, and against what a member is worth. If you don't know those numbers you are negotiating blind in a conversation where they are not.

The shape of it. A gym's economics are retention economics. A member who stays twelve months is worth several times one who leaves in three, the cost of replacing them is high, and

most leaving happens early. Those three facts are the whole reason a product like PODIUM could be worth paying for — you are not selling photographs, you are selling a reason to come back in week six.

What the numbers roughly say, with a serious caveat below:

- Annual retention lands somewhere around **66–71%**, with boutique studios claimed higher ([GymMaster](https://gymmaster.com/blog/gym-membership-and-retention-statistics), [Smart Health Clubs](https://smarthealthclubs.com/blog/everything-you-need-to-know-about-gym-member-retention/))
- **Nearly half of new members leave within six months** — early churn is where the money goes
- Acquiring a member is widely claimed to cost **about five times** what keeping one does
- Member lifetime value estimates sit around **$800–$1,240**, which is roughly R15,000–R23,000 — but see the caveat

The caveat, and it is the real lesson here. Almost every one of those numbers comes from a company that sells gym software, on a blog written to make you buy gym software. The "five times" claim is repeated everywhere and cited to nothing. Treat all of it as **directional, not factual**, and notice how easily a confident number with no source behind it becomes something everybody "knows". You are about to generate numbers of your own. This is what they can turn into.

South African context, also directional: the fitness services market is put at roughly **$1bn and growing ~7% a year**, boutique and functional studios are the fastest-growing segment, and supply clusters in Gauteng, the Western Cape and KZN ([Ken Research](https://www.kenresearch.com/industry-reports/south-africa-fitness-services-market)). The one figure worth carrying into the room: **StatsSA recorded gym fees rising 20.5% between October 2024 and October 2025** ([BusinessTech](https://businesstech.co.za/news/?p=847540)). Gyms are already asking members for more money. That cuts both ways and you should think about which way before you are asked.

Write in the notepad:

1. If a member is worth R15,000 over their life and half leave within six months, what is one prevented cancellation worth? Show the arithmetic.
2. How many prevented cancellations a year would make R1,500 a month obviously worth it? Is that number believable?
3. What else could a gym owner buy for R1,500 a month? List five things. Where does PODIUM rank honestly?
4. Gym fees went up 20.5% in a year. Does that make an owner more or less likely to add a monthly cost? Write both arguments, then pick one.

You've got it when you can explain the gym's side of the deal better than the owner can, and when you notice yourself wanting to check a statistic's source before repeating it.

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3 · One light

Why. PODIUM's entire claim is that two photographs taken weeks apart are comparable, and lighting is where that is won. It is also the difference between a photograph someone posts and one they delete. Right now you own one light and one reflector, which is not a limitation — it is the correct setup, and worth understanding properly rather than copying.

Start here: [The Right Way to Light a Physique](https://fstoppers.com/lighting/right-way-light-physique-901623) on Fstoppers. It is the closest thing to your exact situation: a single strip softbox, mounted high, lighting a whole body. It also flags the failure you will hit — **a hand casting a shadow across the torso for the entire shoot**, unnoticed until afterwards.

Then:

- [Adobe's strip box guide] (https://www.adobe.com/creativecloud/photography/hub/guides/stripbox-lighting) — short, text, explains why 45° gives contour
- [Profoto: five portrait setups with one light](https://profoto.com/se/tips/5-portrait-lighting-setups-with-one-profoto-b10-light) — the same light, five results, which teaches distance and angle faster than any amount of theory
- [B&H on strip softboxes](https://www.bhphotovideo.com/explora/photography/tips-and-solutions/strip-softboxes) and [on softboxes generally] (https://www.bhphotovideo.com/explora/photography/tips-and-solutions/how-to-use-a-softbox)

The one physical law to understand properly: light falls off with the square of distance. Move a light twice as far away and you get a quarter the light — but also a *gentler gradient across the body*, because the difference between the near shoulder and the far one shrinks. That single trade-off explains nearly every lighting decision you will make, including why your softbox is at 1200mm and what moving it costs.

Write in the notepad:

1. Draw the softbox at 1200mm from the mark. Now at 2400mm. What changes about the near shoulder versus the far one, and about the shadow edge?
2. Your softbox spans 500mm to 2100mm off the floor. For a 2100mm person, where does the light fall off fastest, and what does that do to a comparison between a tall member and a short one?
3. What would make a member delete a photo of themselves? List five. How many are lighting?
4. What is the *cheapest* change that would most improve the photographs? Not the best — the cheapest.

You've got it when you can look at any photograph of a person and say roughly where the light was.

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4 · Distance, and what it does to a body

Why. You just spent an evening measuring how far the camera has to be for a person to *fit*. That is only half the question. The other half is what that distance does to how they **look** — and since people are going to judge their own bodies in these photographs, it is not a small thing.

The thing most people get wrong, and it is worth getting right: **focal length does not distort faces. Distance does.** A wide lens looks bad for portraits only because it makes you stand close to fill the frame; the same lens from far away is fine. Standing close makes near features large and far features small — a bigger nose, a narrower jaw, a chest that reads as larger than it is.

[Photography Life](https://photographylife.com/does-focal-length-distort-subjects) makes the case cleanly; [Orms, in Cape Town](https://blog.ormsdirect.co.za/?p=5138), says the same thing and is local. [DIY Photography](https://www.diyphotography.net/?p=66463) shows it with head shots at matched framing — the illustration is worth more than the argument.

Why this is good news for PODIUM. A full-body frame needs you far back anyway. At 2600–2900mm you are already outside the distortion zone, so bodies will read honestly. That is partly luck and worth knowing, because the moment someone suggests "let's put the camera closer and crop", you need to know what it would cost.

Write in the notepad:

1. In your own words, why does standing close make a nose look big? Explain it without using the word "lens".
2. If you moved the camera to 1500mm and cropped to the same framing, what would change about the body? Would a member notice? Would they be able to say why?
3. The output is 4:5. Why does that aspect ratio exist, and what does it cut off that 16:9 would have kept?
4. What would happen to a member's then-and-now pair if someone moved the camera 200mm between visits? Would the registration catch it? Should it?

You've got it when you can predict, before looking, whether a photograph was taken close or far.

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5 · Reading a number honestly

Why. In four weeks you will have about 120 sessions and a set of numbers, and you will want them to say yes. Everybody does. `DECISIONS.md` already protects you from most of the ways that goes wrong — thresholds written before the data, a sealed dashboard, one change at a time — but you wrote those from instinct. Learning the names makes them defensible when somebody with a spreadsheet disagrees with you.

The four ideas, and all four are already in your project:

- **Pre-registration.** Decide what counts as success before you see the data. Why: once you have the numbers, every one of them can be argued into meaning what you were already hoping.
- **Confidence intervals.** At n = 40 a measured 30% return rate is really "somewhere between about 17% and 47%". The interval *is* the finding. Your write-up already prints these.
- **Selection bias.** Your keep rate is measured on people who agreed to take part. That is not the same population as "people in the gym", and the gap is the whole reason the day sheet has walk-away tallies on it.
- **Multiple comparisons.** Look at twenty numbers and one of them will look significant by chance. This is why there are two go/no-go rules and not eleven.

Where to learn it properly: any introductory statistics course covers all four — Khan Academy's statistics and probability track is free and well structured. For the applied version, read about A/B testing pitfalls rather than textbook statistics; the failure modes are described in plainer language and they are the ones you will actually hit.

Write in the notepad:

1. Explain a confidence interval to someone who has never heard of one. Two sentences, no jargon.
2. Your pilot will tell you 30% from 10%. It cannot tell you 30% from 22%. Why not? And what would you need to tell them apart?
3. Name three ways your own pilot could produce a flattering number that isn't true. Be specific to PODIUM, not general.
4. If the four weeks come back ambiguous, what will you actually do? Write it down **now**, while it costs nothing to be honest.

You've got it when question 4 has an answer you would still stand behind after seeing data you didn't like.

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6 · OpenSCAD

Why. You are doing the pod yourself, and the AI writing the code is going to be confidently wrong sometimes. You do not need to be fluent — you need enough to read what it hands you and notice when a number has quietly changed.

Start here: the official OpenSCAD cheat sheet — one page, every command. Find it from **openscad.org → Documentation → Cheat Sheet**; I could not open openscad.org from where I am running to give you the exact link, so navigate rather than trusting a URL I typed from memory. A [mirrored copy with commentary](https://openhome.cc/eGossip/OpenSCAD/CheatSheet.html) works if the original is down. Then [Materialise's beginner tutorial](https://i.materialise.com/blog/?p=21410), which has four short videos and explains the core idea: everything is simple shapes added to and subtracted from each other.

The one concept that trips everyone up, and it will trip you: **OpenSCAD is not a normal programming language.** Variables are not assignments that happen in order — they are set once for the whole scope, so writing `x = 1;` and then `x = 2;` gives you 2 everywhere, including on the line above. [This explanation](https://openhome.cc/eGossip/OpenSCAD/CheatSheet.html) covers it. Knowing this saves you an hour of wondering why your model ignores you.

Write in the notepad:

1. The five primitives and the three boolean operations, from memory, with a sketch of each. (Yes, a sketch — stick to boxes and circles, nobody is judging.)
2. What is the difference between F5 and F6, and why does it matter on your laptop?
3. What does `%` do to a shape, and why is that exactly right for the camera's field-of-view cone?
4. Write down the parameter block from the brief, by hand, with its units. If you can write it out, you will notice when the AI changes one.

You've got it when you can read an AI-written SCAD file and say what it builds before you press F5.

If you only do two things

Read *The Mom Test* tonight, and rewrite your six questions. That is the one that unblocks everything else, and it is an evening.

Then start the "questions I cannot answer yet" page. Not as an exercise — as a habit. In four weeks the pilot will answer three of them, the gym owner will answer two, and the rest will tell you what to build next.

One warning, free

Every topic above has an infinite amount of material behind it, and reading is much more comfortable than measuring a corner or phoning a fabricator.

You already have the three highest-value things on the list, and none of them are reading: **measure the camera, tape the footprint, talk to the owner.** Reading makes those conversations better. It does not replace them, and if the notepad fills up faster than the field-research sheet does, that is the thing to notice.